

Configure ASA Basic Settings and Firewall Using the CLI

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	209.165.200.225	255.255.255.248	N/A
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	
R3	G0/1	172.16.3.1	255.255.255.0	N/A
	S0/0/1	10.2.2.1	255.255.255.252	
ASA	G1/1	209.165.200.226	255.255.255.248	NA
	G1/2	192.168.1.1	255.255.255.0	
	G1/3	192.168.2.1	255.255.255.0	
DMZ Server	NIC	192.168.2.3	255.255.255.0	192.168.2.1
PC-B	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C	NIC	172.16.3.3	255.255.255.0	172.16.3.1

Objectives

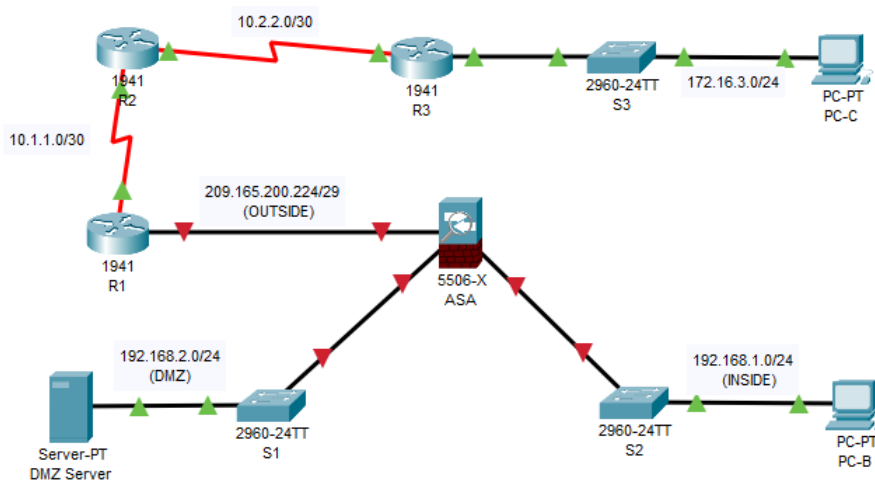
- Verify connectivity and explore the ASA
- Configure basic ASA settings and interface security levels using CLI
- Configure routing, address translation, and inspection policy using CLI
- Configure DHCP, AAA, and SSH
- Configure a DMZ, Static NAT, and ACLs

Scenario

Your company has one location connected to an ISP. R1 represents a CPE device managed by the ISP. R2 represents an intermediate Internet router. R3 represents an ISP that connects an administrator from a network management company, who has been hired to remotely manage your network. The ASA is an edge CPE security device that connects the internal corporate network and DMZ to the ISP while providing NAT and DHCP services to inside hosts. The ASA will be configured for management by an administrator on the internal network and by the remote administrator. The ISP assigned the public IP address space of 209.165.200.224/29, which will be used for address translation on the ASA.

All router and switch devices have been preconfigured with the following:

- Enable password: **ciscoenpa55**
- Console password: **ciscoconpa55**
- Admin username and password: **admin/adminpa55**



Step 1: Verify connectivity.

The ASA is not currently configured. However, all routers, PCs, and the DMZ server are configured. Verify that PC-C can ping any router interface. PC-C is unable to ping the ASA, PC-B, or the DMZ server.

Physical Config Desktop Programming Attributes

Command Prompt X

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 172.16.3.1

Pinging 172.16.3.1 with 32 bytes of data:

Reply from 172.16.3.1: bytes=32 time<1ms TTL=255

Reply from 172.16.3.1: bytes=32 time<1ms TTL=255

Reply from 172.16.3.1: bytes=32 time<1ms TTL=255

Reply from 172.16.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.3.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.1.1.2

Pinging 10.1.1.2 with 32 bytes of data:

Reply from 10.1.1.2: bytes=32 time=1ms TTL=254

Reply from 10.1.1.2: bytes=32 time=1ms TTL=254

Reply from 10.1.1.2: bytes=32 time=1ms TTL=254

Reply from 10.1.1.2: bytes=32 time=1ms TTL=254

Ping statistics for 10.1.1.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 10.1.1.1

Pinging 10.1.1.1 with 32 bytes of data:

Reply from 10.1.1.1: bytes=32 time=3ms TTL=253

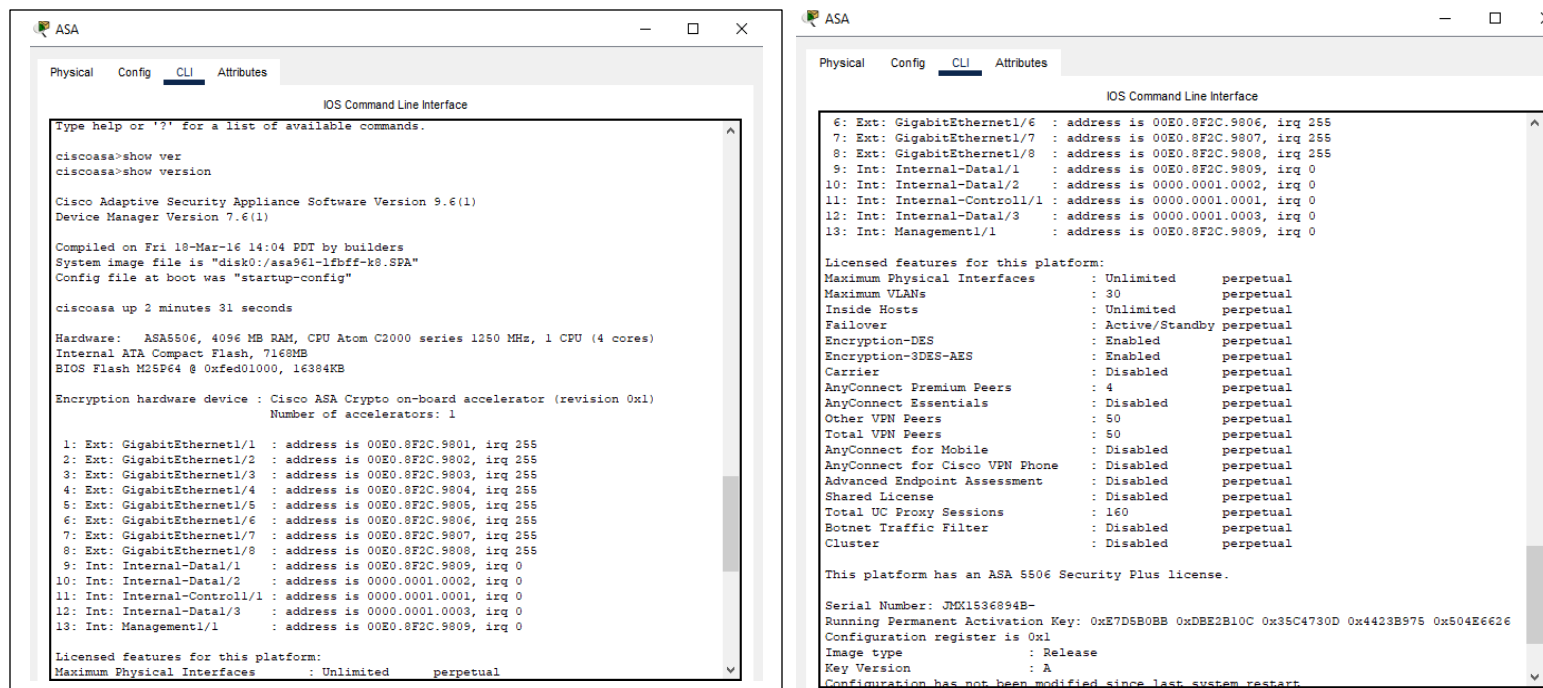
Reply from 10.1.1.1: bytes=32 time=2ms TTL=253

Reply from 10.1.1.1: bytes=32 time=2ms TTL=253

Reply from 10.1.1.1: bytes=32 time=3ms TTL=253

Step 2: Determine the ASA version, interfaces, and license.

Use the **show version** command to determine various aspects of this ASA device.



The first screenshot shows the output of the 'show version' command in the ASA CLI. It displays the software version (9.6(1)), hardware details (ASA5506, 4096 MB RAM, CPU Atom C2000 series), and a list of interfaces with their addresses and IRQs. The second screenshot shows the output of the 'show version' command, displaying the licensed features for the platform, including Maximum Physical Interfaces, Maximum VLANs, Inside Hosts, Failover, Encryption-DES, Encryption-3DES-AES, Carrier, AnyConnect Premium Peers, AnyConnect Essentials, Other VPN Peers, Total VPN Peers, AnyConnect for Mobile, AnyConnect for Cisco VPN Phone, Advanced Endpoint Assessment, Shared License, Total UC Proxy Sessions, Botnet Traffic Filter, and Cluster. It also shows the serial number, running permanent activation key, configuration register, image type, key version, and a message indicating that the configuration has not been modified since the last system restart.

```
ASA
Physical Config CLI Attributes
IOS Command Line Interface
Type help or '?' for a list of available commands.
ciscoasa>show ver
ciscoasa>show version

Cisco Adaptive Security Appliance Software Version 9.6(1)
Device Manager Version 7.6(1)

Compiled on Fri 18-Mar-16 14:04 PDT by builders
System image file is "disk0:/asa961-lfbff-k8.SPA"
Config file at boot was "startup-config"

ciscoasa up 2 minutes 31 seconds

Hardware: ASA5506, 4096 MB RAM, CPU Atom C2000 series 1250 MHz, 1 CPU (4 cores)
Internal ATA Compact Flash, 7168MB
BIOS Flash M25P64 @ 0xfed01000, 16384KB

Encryption hardware device : Cisco ASA Crypto on-board accelerator (revision 0x1)
Number of accelerators: 1

1: Ext: GigabitEthernet1/1 : address is 00E0.8F2C.9801, irq 255
2: Ext: GigabitEthernet1/2 : address is 00E0.8F2C.9802, irq 255
3: Ext: GigabitEthernet1/3 : address is 00E0.8F2C.9803, irq 255
4: Ext: GigabitEthernet1/4 : address is 00E0.8F2C.9804, irq 255
5: Ext: GigabitEthernet1/5 : address is 00E0.8F2C.9805, irq 255
6: Ext: GigabitEthernet1/6 : address is 00E0.8F2C.9806, irq 255
7: Ext: GigabitEthernet1/7 : address is 00E0.8F2C.9807, irq 255
8: Ext: GigabitEthernet1/8 : address is 00E0.8F2C.9808, irq 255
9: Int: Internal-Datal/1 : address is 00E0.8F2C.9809, irq 0
10: Int: Internal-Datal/2 : address is 0000.0001.0002, irq 0
11: Int: Internal-Controll/1 : address is 0000.0001.0001, irq 0
12: Int: Internal-Datal/3 : address is 0000.0001.0003, irq 0
13: Int: Management1/1 : address is 00E0.8F2C.9809, irq 0

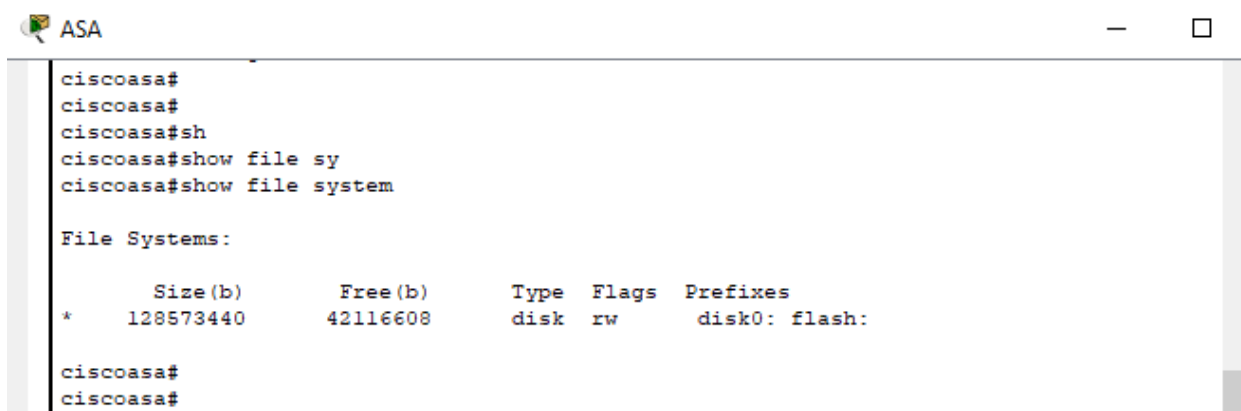
Licensed features for this platform:
Maximum Physical Interfaces : Unlimited perpetual
Maximum VLANs : 30 perpetual
Inside Hosts : Unlimited perpetual
Failover : Active/Standby perpetual
Encryption-DES : Enabled perpetual
Encryption-3DES-AES : Enabled perpetual
Carrier : Disabled perpetual
AnyConnect Premium Peers : 4 perpetual
AnyConnect Essentials : Disabled perpetual
Other VPN Peers : 50 perpetual
Total VPN Peers : 50 perpetual
AnyConnect for Mobile : Disabled perpetual
AnyConnect for Cisco VPN Phone : Disabled perpetual
Advanced Endpoint Assessment : Disabled perpetual
Shared License : Disabled perpetual
Total UC Proxy Sessions : 160 perpetual
Botnet Traffic Filter : Disabled perpetual
Cluster : Disabled perpetual

This platform has an ASA 5506 Security Plus license.

Serial Number: JMK1S36894B-
Running Permanent Activation Key: 0xE7D5B0BB 0xDBE2B10C 0x35C4730D 0x4423B975 0x504E6626
Configuration register is 0x1
Image type : Release
Key Version : A
Configuration has not been modified since last system restart.
```

Step 3: Determine the file system and contents of flash memory.

- Enter privileged EXEC mode. A password has not been set. Press **Enter** when prompted for a password.
- Use the **show file system** command to display the ASA file system and determine which prefixes are supported.



The screenshot shows the output of the 'show file system' command in the ASA CLI. It displays the file systems available on the device, including the disk0: flash: file system. The output shows the size of the file system (128573440 bytes) and the free space (42116608 bytes). The file system is a disk with read-write (rw) permissions. The output also shows the prefixes supported by the file system, which are disk0: and flash:.

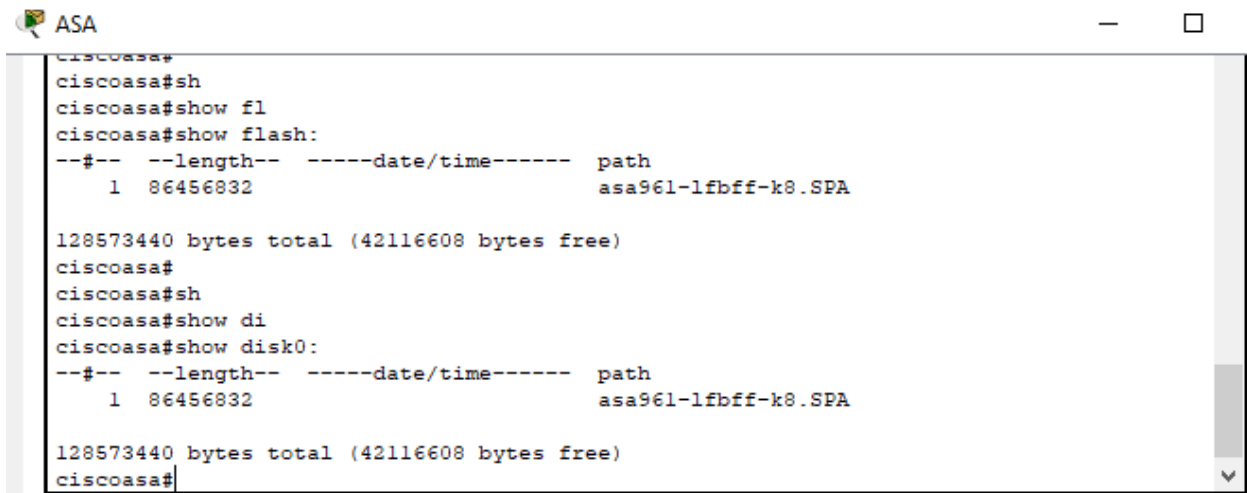
```
ASA
Physical Config CLI Attributes
IOS Command Line Interface
ciscoasa#
ciscoasa#
ciscoasa#sh
ciscoasa#show file sy
ciscoasa#show file system

File Systems:

      Size(b)      Free(b)      Type  Flags  Prefixes
*    128573440    42116608    disk  rw     disk0: flash:

ciscoasa#
ciscoasa#
```

- c. Use the **show flash:** or **show disk0:** command to display the contents of flash memory.



```
ASA
ciscoasa#
ciscoasa#sh
ciscoasa#show fl
ciscoasa#show flash:
--#-- --length-- -----date/time----- path
      1 86456832                               asa961-1fbff-k8.SPA

128573440 bytes total (42116608 bytes free)
ciscoasa#
ciscoasa#sh
ciscoasa#show di
ciscoasa#show disk0:
--#-- --length-- -----date/time----- path
      1 86456832                               asa961-1fbff-k8.SPA

128573440 bytes total (42116608 bytes free)
ciscoasa#
```

Part 2: Configure ASA Settings and Interface Security Using the CLI

Step 1: Configure the hostname and domain name.

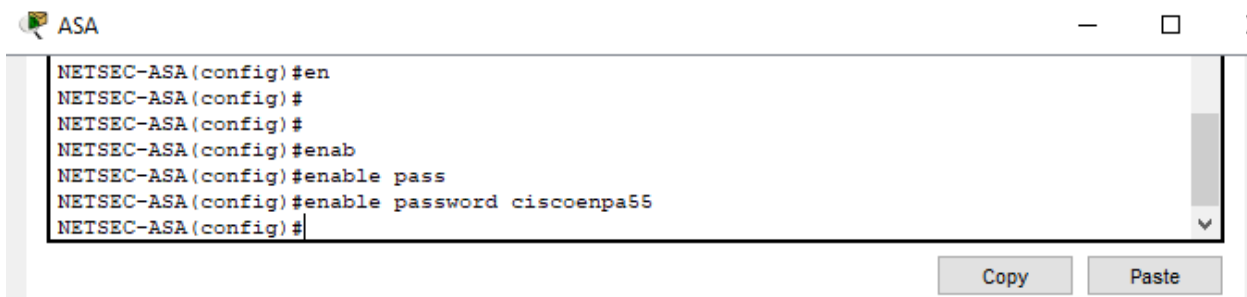
- Configure the ASA hostname as **NETSEC-ASA**.
- Configure the domain name as **netsec.com**.



```
ASA
ciscoasa#configure terminal
ciscoasa(config)#
ciscoasa(config)#ho
ciscoasa(config)#hostname NETSEC-ASA
NETSEC-ASA(config)#do
NETSEC-ASA(config)#domain-name netsec.com
NETSEC-ASA(config)#
```

Step 2: Configure the enable mode password.

Use the **enable password** command to change the privileged EXEC mode password to **ciscoenpa55**.



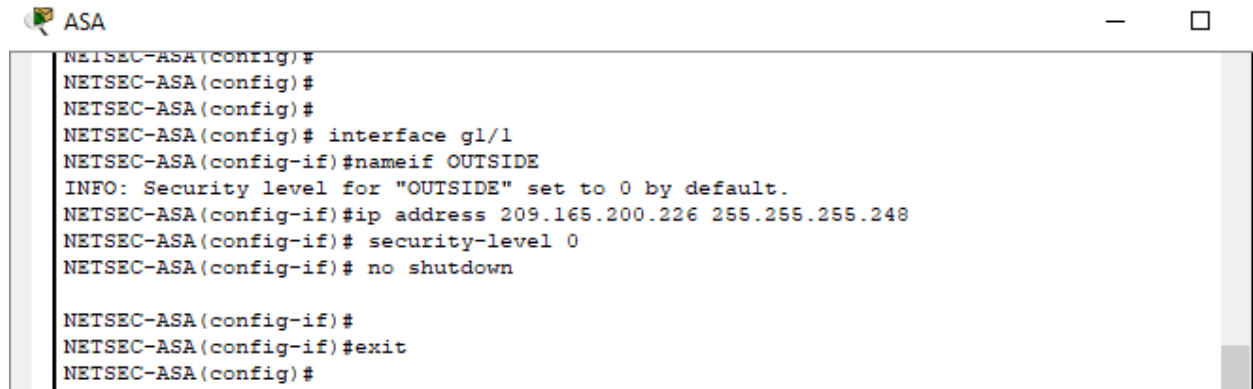
```
ASA
NETSEC-ASA(config)#en
NETSEC-ASA(config)#
NETSEC-ASA(config)#
NETSEC-ASA(config)#enab
NETSEC-ASA(config)#enable pass
NETSEC-ASA(config)#enable password ciscoenpa55
NETSEC-ASA(config)#
```

Copy Paste

Step 3: Configure the INSIDE and OUTSIDE interfaces.

You will only configure the G1/1 (OUTSIDE) and G1/2 (INSIDE) interfaces at this time. The G1/3 (DMZ) interface will be configured in Part 5 of the activity.

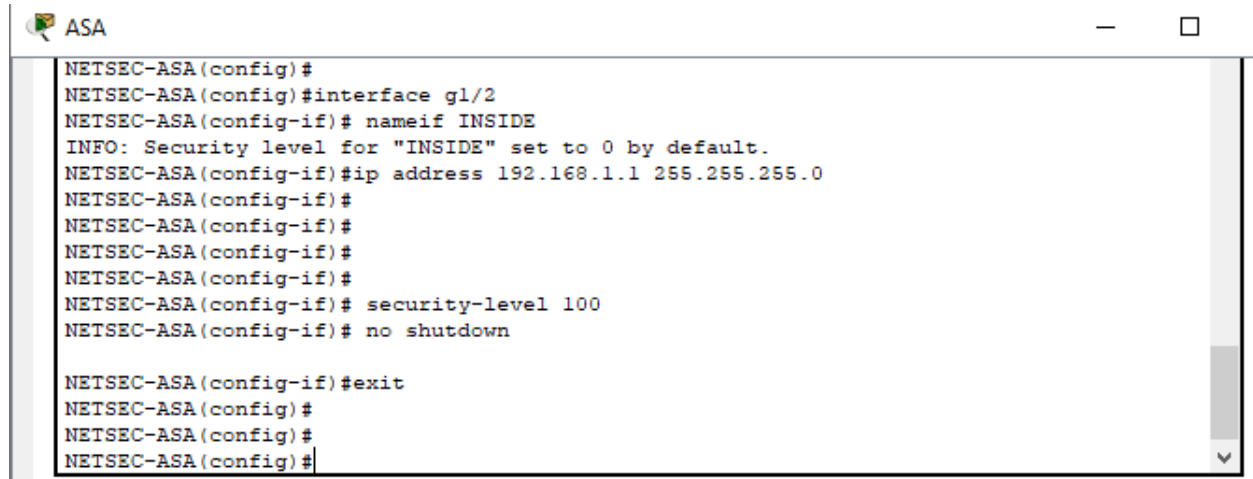
- a. Create the G1/1 interface for the outside network (209.165.200.224/29), set the security level to the lowest setting of 0, and enable the interface.



```
ASA
NETSEC-ASA(config)#
NETSEC-ASA(config)#
NETSEC-ASA(config)#
NETSEC-ASA(config)# interface g1/1
NETSEC-ASA(config-if)#nameif OUTSIDE
INFO: Security level for "OUTSIDE" set to 0 by default.
NETSEC-ASA(config-if)#ip address 209.165.200.226 255.255.255.248
NETSEC-ASA(config-if)# security-level 0
NETSEC-ASA(config-if)# no shutdown

NETSEC-ASA(config-if)#
NETSEC-ASA(config-if)#exit
NETSEC-ASA(config)#
```

- b. Configure the G1/2 interface for the inside network (192.168.1.0/24) and set the security level to the highest setting of 100 and enable the interface.



```
ASA
NETSEC-ASA(config)#
NETSEC-ASA(config)#interface g1/2
NETSEC-ASA(config-if)# nameif INSIDE
INFO: Security level for "INSIDE" set to 0 by default.
NETSEC-ASA(config-if)#ip address 192.168.1.1 255.255.255.0
NETSEC-ASA(config-if)#
NETSEC-ASA(config-if)#
NETSEC-ASA(config-if)#
NETSEC-ASA(config-if)# security-level 100
NETSEC-ASA(config-if)# no shutdown

NETSEC-ASA(config-if)#exit
NETSEC-ASA(config)#
NETSEC-ASA(config)#
NETSEC-ASA(config)#
```

- c. Use the following verification commands to check your configurations:
 - 1) Use the **show interface ip brief** command to display the status for all ASA interfaces.

```

NETSEC-ASA#show in
NETSEC-ASA#show interface ip
NETSEC-ASA#show interface ip br
NETSEC-ASA#show interface ip brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Virtual0	127.1.0.1	YES	unset	up	up
GigabitEthernet1/1	209.165.200.226	YES	manual	up	up
GigabitEthernet1/2	192.168.1.1	YES	manual	up	up
GigabitEthernet1/3	unassigned	YES	unset	administratively down	down
GigabitEthernet1/4	unassigned	YES	unset	administratively down	down
GigabitEthernet1/5	unassigned	YES	unset	administratively down	down
GigabitEthernet1/6	unassigned	YES	unset	administratively down	down
GigabitEthernet1/7	unassigned	YES	unset	administratively down	down
GigabitEthernet1/8	unassigned	YES	unset	administratively down	down
Management1/1	unassigned	YES	unset	administratively down	down
Internal-Controll/1	127.0.1.1	YES	unset	up	up
Internal-Datal/1	unassigned	YES	unset	up	up
Internal-Datal/2	unassigned	YES	unset	up	up
Internal-Datal/3	unassigned	YES	unset	up	up

```

NETSEC-ASA#
NETSEC-ASA#
NETSEC-ASA#
NETSEC-ASA#

```

2) Use the **show ip address** command to display the interface information.

IOS Command Line Interface

```

NETSEC-ASA#sh
NETSEC-ASA#show ip add
NETSEC-ASA#show ip address

```

System IP Addresses:

Interface	Name	IP address	Subnet mask	Method
GigabitEthernet1/1	OUTSIDE	209.165.200.226	255.255.255.248	manual
GigabitEthernet1/2	INSIDE	192.168.1.1	255.255.255.0	manual
GigabitEthernet1/3		unassigned	unassigned	unset
GigabitEthernet1/4		unassigned	unassigned	unset
GigabitEthernet1/5		unassigned	unassigned	unset
GigabitEthernet1/6		unassigned	unassigned	unset
GigabitEthernet1/7		unassigned	unassigned	unset
GigabitEthernet1/8		unassigned	unassigned	unset
Management1/1		unassigned	unassigned	unset

Current IP Addresses:

Interface	Name	IP address	Subnet mask	Method
GigabitEthernet1/1	OUTSIDE	209.165.200.226	255.255.255.248	manual
GigabitEthernet1/2	INSIDE	192.168.1.1	255.255.255.0	manual
GigabitEthernet1/3		unassigned	unassigned	unset
GigabitEthernet1/4		unassigned	unassigned	unset
GigabitEthernet1/5		unassigned	unassigned	unset
GigabitEthernet1/6		unassigned	unassigned	unset
GigabitEthernet1/7		unassigned	unassigned	unset
GigabitEthernet1/8		unassigned	unassigned	unset
Management1/1		unassigned	unassigned	unset

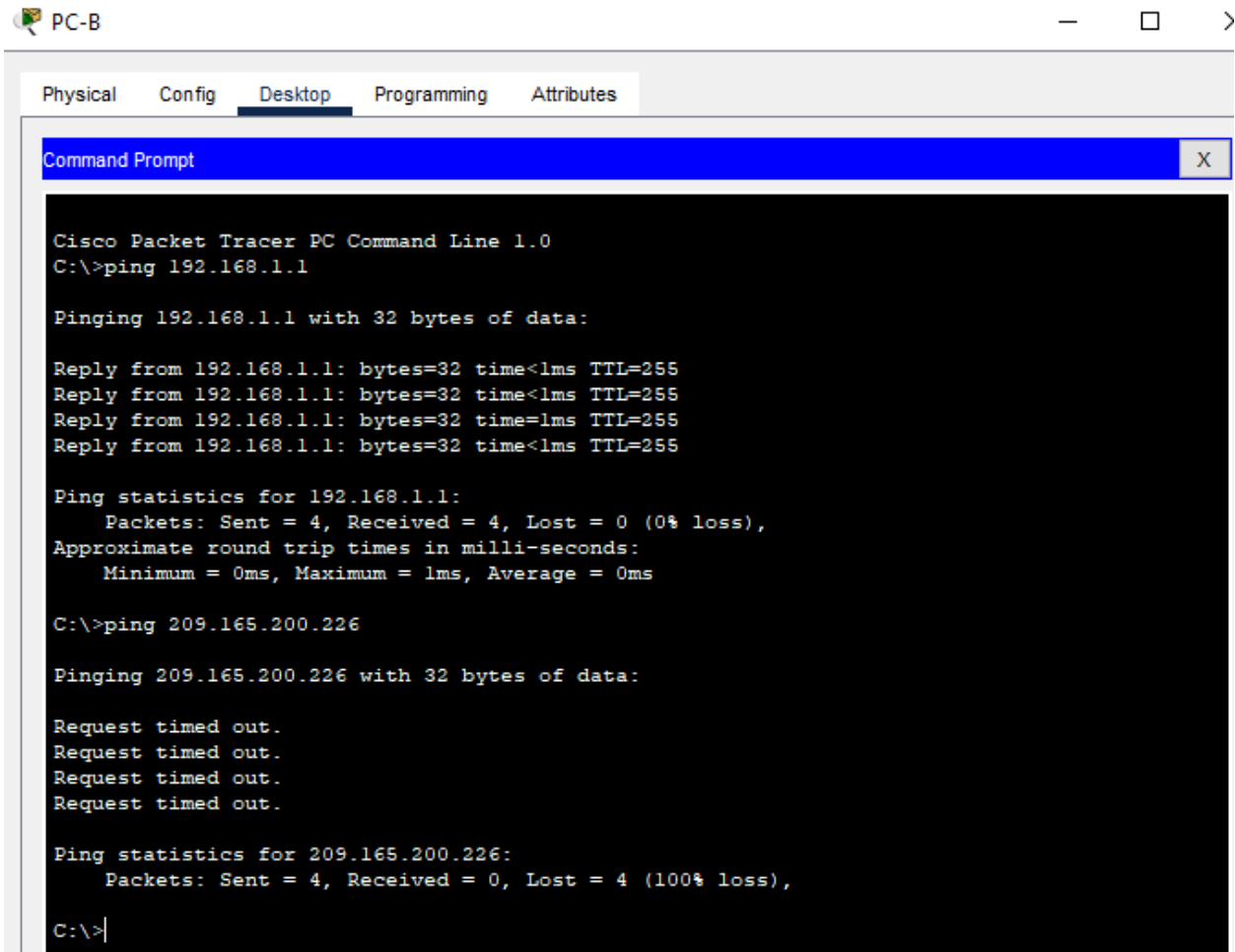
```

NETSEC-ASA#

```

Step 4: Test connectivity to the ASA.

- a. You should be able to ping from PC-B to the ASA inside interface address (192.168.1.1). If the pings fail, troubleshoot the configuration as necessary.
- b. From PC-B, ping the G1/1 (OUTSIDE) interface at IP address 209.165.200.226. You should not be able to ping this address.



The screenshot shows a Packet Tracer window for PC-B. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the results of two ping commands. The first command, 'ping 192.168.1.1', is successful, showing four replies with 32 bytes of data, a time of less than 1ms, and a TTL of 255. The statistics show 4 packets sent, 4 received, and 0 lost (0% loss). The second command, 'ping 209.165.200.226', fails, showing four 'Request timed out' messages. The statistics show 4 packets sent, 0 received, and 4 lost (100% loss).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 209.165.200.226

Pinging 209.165.200.226 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 209.165.200.226:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

Part 3: Configure Routing, Address Translation, and Inspection Policy Using the CLI

Step 1: Configure a static default route for the ASA.

Configure a default static route on the ASA OUTSIDE interface to enable the ASA to reach external networks.

- Create a “quad zero” default route using the **route** command, associate it with the ASA OUTSIDE interface, and point to the R1 G0/0 IP address (209.165.200.225) as the gateway of last resort.

```
NETSEC-ASA#
NETSEC-ASA#conf t
NETSEC-ASA(config)#route OUTSIDE 0.0.0.0 0.0.0.0 209.165.200.225
NETSEC-ASA(config)#
NETSEC-ASA(config)#
NETSEC-ASA(config)#
```

- Issue the **show route** command to verify the static default route is in the ASA routing table.

```
NETSEC-ASA#show ro
NETSEC-ASA#show route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 209.165.200.225 to network 0.0.0.0

C    192.168.1.0 255.255.255.0 is directly connected, INSIDE, GigabitEthernet1/2
    209.165.200.0/29 is subnetted, 2 subnets
C      209.165.200.0 255.255.255.248 is directly connected, OUTSIDE,
GigabitEthernet1/1
C      209.165.200.224 255.255.255.248 is directly connected, OUTSIDE,
GigabitEthernet1/1
S*   0.0.0.0/0 [1/0] via 209.165.200.225
NETSEC-ASA#
```

- Verify that the ASA can ping the R1 S0/0/0 IP address 10.1.1.1. If the ping is unsuccessful, troubleshoot as necessary.

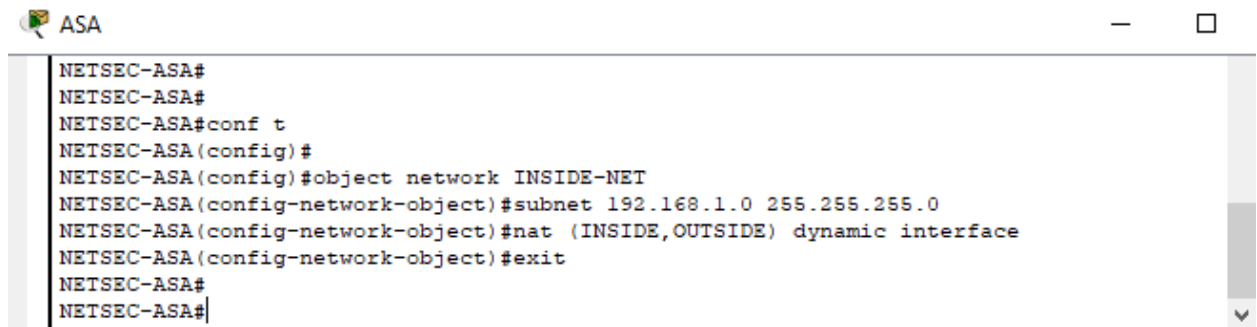
```
NETSEC-ASA#
NETSEC-ASA#
NETSEC-ASA#
NETSEC-ASA#ping 10.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

NETSEC-ASA#
```

Step 2: Configure address translation using PAT and network objects.

- a. Create network object **INSIDE-NET** and assign attributes to it using the **subnet** and **nat** commands.

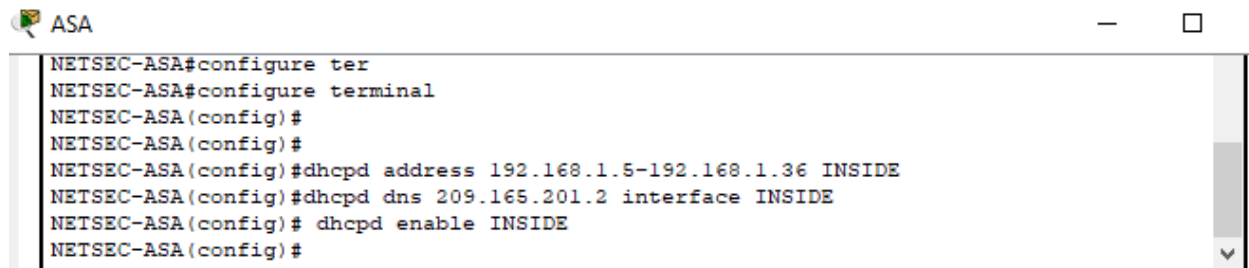
A terminal window titled "ASA" with a standard Linux-style title bar (minimize, maximize, close buttons). The terminal shows the following commands and output:

```
NETSEC-ASA#  
NETSEC-ASA#  
NETSEC-ASA#conf t  
NETSEC-ASA(config)#  
NETSEC-ASA(config)#object network INSIDE-NET  
NETSEC-ASA(config-network-object)#subnet 192.168.1.0 255.255.255.0  
NETSEC-ASA(config-network-object)#nat (INSIDE,OUTSIDE) dynamic interface  
NETSEC-ASA(config-network-object)#exit  
NETSEC-ASA#  
NETSEC-ASA#
```

Part 4: Configure DHCP, AAA, and SSH

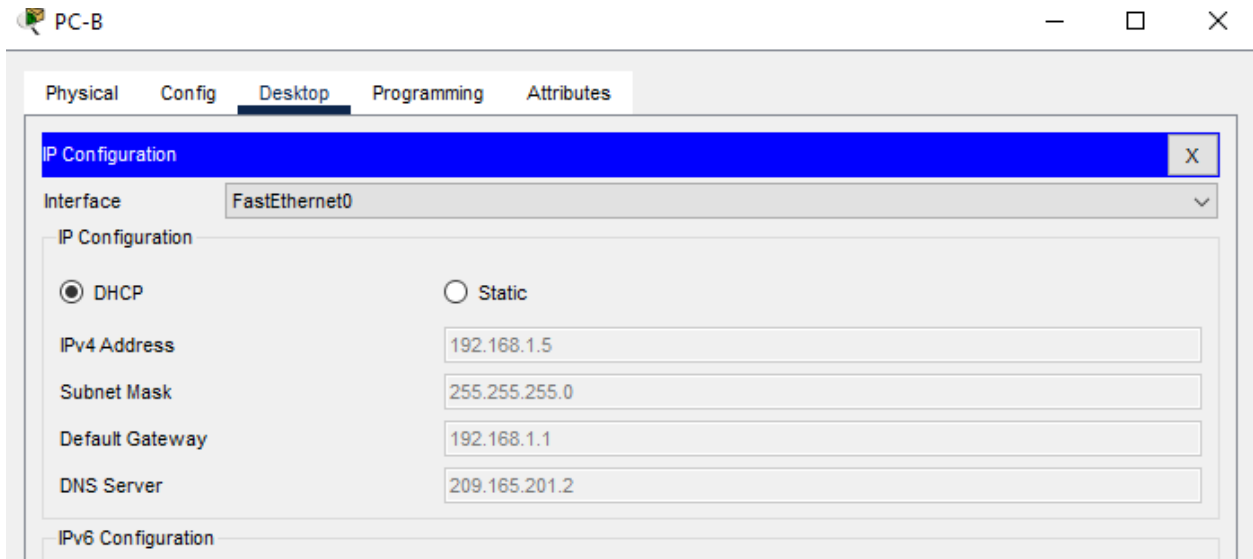
Step 1: Configure the ASA as a DHCP server.

- a. Configure a DHCP address pool and enable it on the ASA INSIDE interface.
- b. (Optional) Specify the IP address of the DNS server to be given to clients.
- c. Enable the DHCP daemon within the ASA to listen for DHCP client requests on the enabled interface (INSIDE).

A terminal window titled "ASA" with a standard Linux-style title bar. The terminal shows the following commands and output:

```
NETSEC-ASA#configure ter  
NETSEC-ASA#configure terminal  
NETSEC-ASA(config)#  
NETSEC-ASA(config)#  
NETSEC-ASA(config)#dhcpd address 192.168.1.5-192.168.1.36 INSIDE  
NETSEC-ASA(config)#dhcpd dns 209.165.201.2 interface INSIDE  
NETSEC-ASA(config)# dhcpd enable INSIDE  
NETSEC-ASA(config)#
```

- d. Change PC-B from a static IP address to a DHCP client and verify that it receives IP addressing information. Troubleshoot, as necessary to resolve any problems.



Step 2: Configure AAA to use the local database for authentication.

- Define a local user named **admin** by entering the **username** command. Specify a password of **adminpa55**.
- Configure AAA to use the local ASA database for SSH user authentication.

Step 3: Configure remote access to the ASA.

The ASA can be configured to accept connections from a single host or a range of hosts on the INSIDE or OUTSIDE network. In this step, hosts from the OUTSIDE network can only use SSH to communicate with the ASA. SSH sessions can be used to access the ASA from the inside network.

- Generate an RSA key pair, which is required to support SSH connections. Because the ASA device has RSA keys already in place, enter **no** when prompted to replace them.

```
NETSEC-ASA(config)#username admin password adminpa55
NETSEC-ASA(config)#aaa authentication ssh console LOCAL
NETSEC-ASA(config)#
NETSEC-ASA(config)#crypto key generate rsa modulus 1024
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.

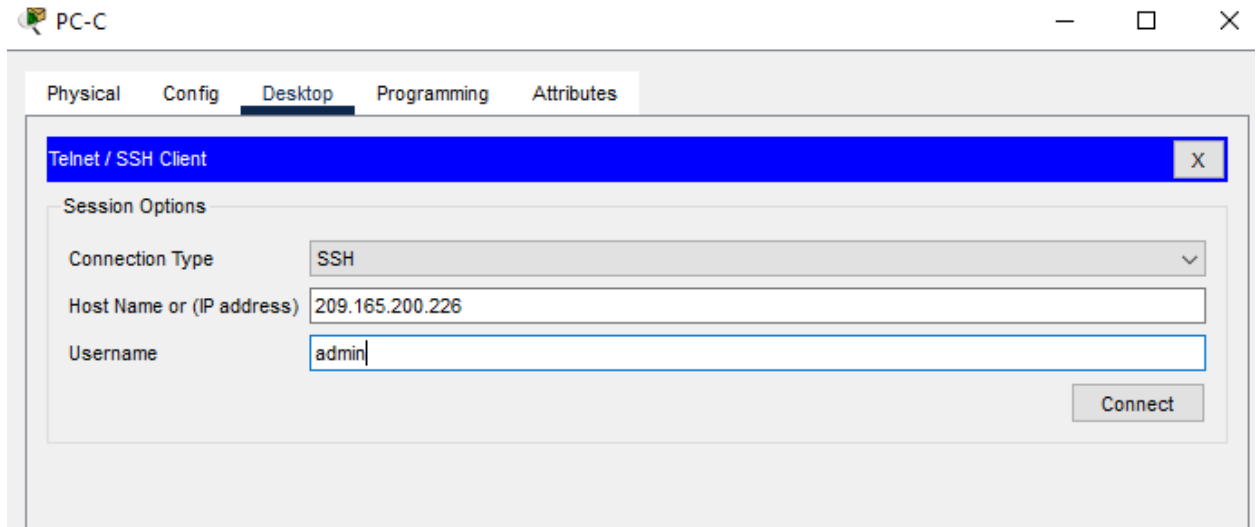
Do you really want to replace them? [yes/no]: no
ERROR: Failed to create new RSA keys named <Default-RSA-Key>
NETSEC-ASA(config)#
NETSEC-ASA(config)#
```

- Configure the ASA to allow SSH connections from any host on the INSIDE network (192.168.1.0/24) and from the remote management host at the branch office (172.16.3.3) on the OUTSIDE network. Set the SSH timeout to 10 minutes (the default is 5 minutes).

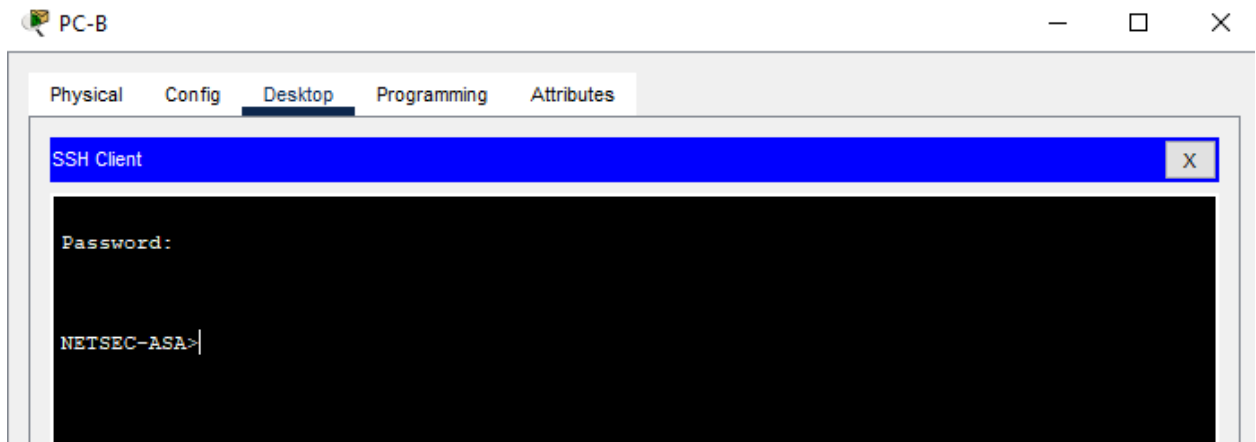
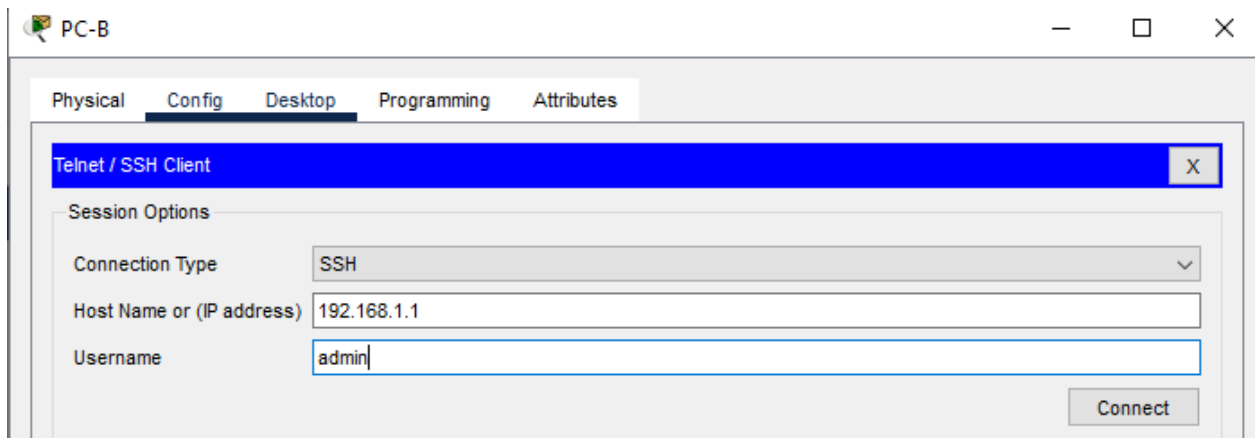
```
NETSEC-ASA(config)#
NETSEC-ASA(config)#ssh 192.168.1.0 255.255.255.0 INSIDE
NETSEC-ASA(config)#ssh 172.16.3.3 255.255.255.255 OUTSIDE
NETSEC-ASA(config)#ssh timeout 10
NETSEC-ASA(config)#
NETSEC-ASA(config)#
```

- c. Establish an SSH session from PC-C to the ASA (209.165.200.226). Troubleshoot if it is not successful.

```
C:\> ssh -l admin 209.165.200.226
```



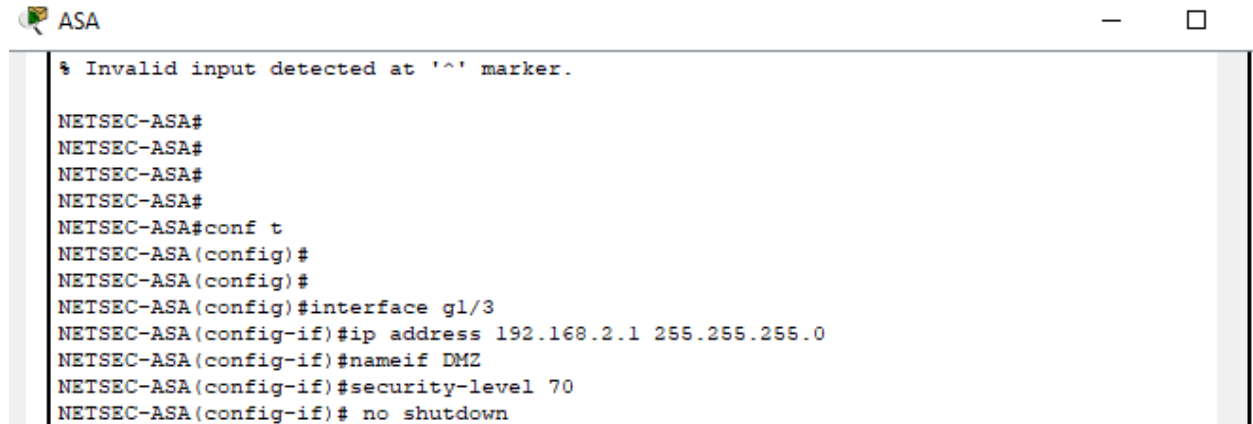
- d. Establish an SSH session from PC-B to the ASA (192.168.1.1). Troubleshoot if it is not successful.



Part 5: Configure a DMZ, Static NAT, and ACLs

Step 1: Configure the DMZ interface VLAN 3 on the ASA.

- Configure DMZ VLAN 3, which is where the public access web server will reside. Assign it IP address 192.168.2.1/24, name it **DMZ**, and assign it a security level of 70. Because the server does not need to initiate communication with the inside users, disable forwarding to interface VLAN 1.

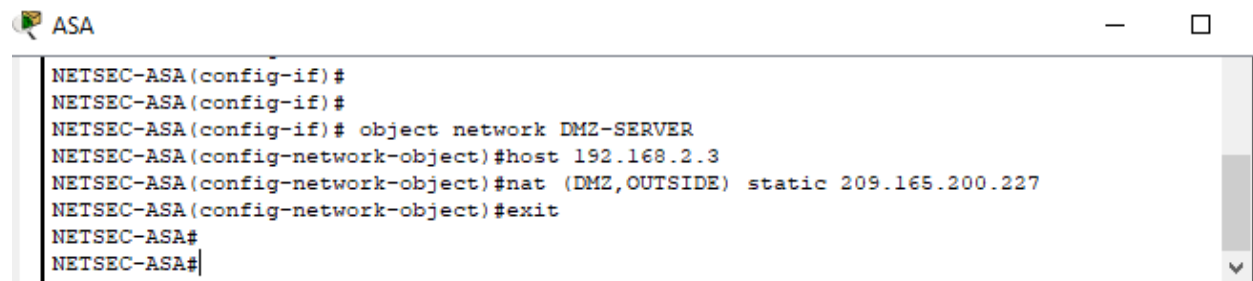


```
ASA
% Invalid input detected at '^' marker.

NETSEC-ASA#
NETSEC-ASA#
NETSEC-ASA#
NETSEC-ASA#
NETSEC-ASA#conf t
NETSEC-ASA(config)#
NETSEC-ASA(config)#
NETSEC-ASA(config)#interface g1/3
NETSEC-ASA(config-if)#ip address 192.168.2.1 255.255.255.0
NETSEC-ASA(config-if)#nameif DMZ
NETSEC-ASA(config-if)#security-level 70
NETSEC-ASA(config-if)# no shutdown
```

Step 2: Configure static NAT to the DMZ server using a network object.

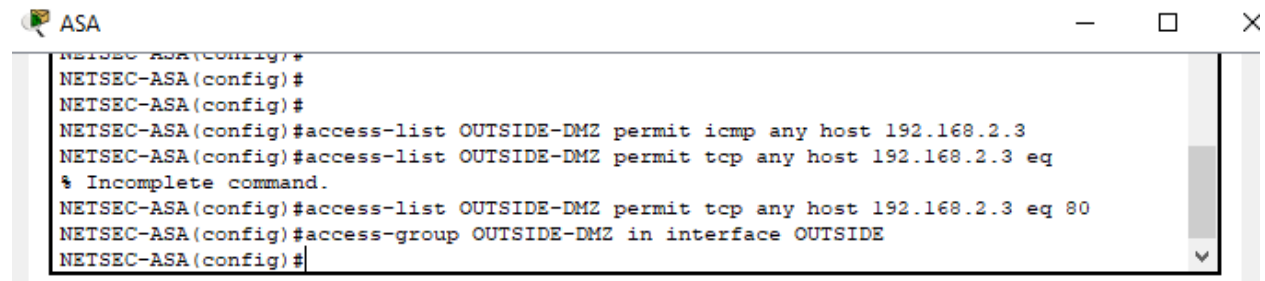
Configure a network object named **DMZ-SERVER** and assign it the static IP address of the DMZ server (192.168.2.3). While in object definition mode, use the **nat** command to specify that this object is used to translate a DMZ address to an OUTSIDE address using static NAT, and specify a public translated address of 209.165.200.227.



```
ASA
NETSEC-ASA(config-if)#
NETSEC-ASA(config-if)#
NETSEC-ASA(config-if)# object network DMZ-SERVER
NETSEC-ASA(config-network-object)#host 192.168.2.3
NETSEC-ASA(config-network-object)#nat (DMZ,OUTSIDE) static 209.165.200.227
NETSEC-ASA(config-network-object)#exit
NETSEC-ASA#
NETSEC-ASA#
```

Step 3: Configure an ACL to allow access to the DMZ server from the Internet.

Configure a named access list **OUTSIDE-DMZ** that permits the TCP protocol on port 80 from any external host to the internal IP address of the DMZ server. Apply the access list to the ASA OUTSIDE interface in the "IN" direction.



```
ASA
NETSEC-ASA(config)#
NETSEC-ASA(config)#
NETSEC-ASA(config)#access-list OUTSIDE-DMZ permit icmp any host 192.168.2.3
NETSEC-ASA(config)#access-list OUTSIDE-DMZ permit tcp any host 192.168.2.3 eq
% Incomplete command.
NETSEC-ASA(config)#access-list OUTSIDE-DMZ permit tcp any host 192.168.2.3 eq 80
NETSEC-ASA(config)#access-group OUTSIDE-DMZ in interface OUTSIDE
NETSEC-ASA(config)#
```

Step 4: Test access to the DMZ server.

From a web browser on PC-C, navigate to the DMZ server (209.165.200.227). Troubleshoot if it is not successful.

